



A new sensor alerts engineers when newly laid concrete pavement is ready for use

Researchers at Purdue University have developed a sensor that enables concrete to communicate, reduces construction time, and the need for frequent pavement maintenance.

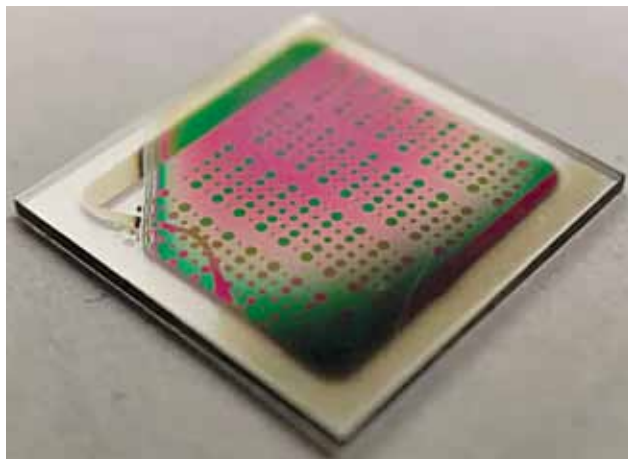


Sensors developed by Luna Lu and her team are installed into the formwork of Interstate 35 in Texas (Credit: Luna Lu)

nance. The sensor alerts engineers via a smartphone app, indicating when the pavement is ready for heavy traffic. This reduces repairs and enables timely road openings. Workers install sensors in concrete formwork and connect to a handheld device for data logging. The app provides real-time concrete strength updates. The technology minimises traffic delays, reducing carbon emissions from road repairs. Researchers aim to lower cement usage in concrete mixes, using AI and sensor data to optimise designs and curb carbon emissions.

New oxygen-ion battery can replace lithium-ion for large-scale energy storage

Researchers at TU Wien have developed a durable oxygen-ion battery, a promising alternative to lithium-ion with a long lifespan and no irreversible capacity loss. The ceramic materials absorb and release double negative oxygen ions, enabling ion migration and generating electric current. Ceramic batteries resemble lithium-ion batteries but offer non-flammability and avoid rare and harmful elements. The key advantage of the new battery is its extended lifespan. Unlike conventional batteries, it can regenerate oxygen and overcome the capacity loss. While it has lower energy density and requires higher temperatures, it shows promise for energy storage. The researchers believe the



A prototype of the battery at TU Wien (Credit: TU Wien)

oxygen-ion battery is ideal for large-scale energy storage such as solar or wind power. Its long lifespan, non-flammable nature, and lack of rare elements in storage buildings outweigh lower energy density and higher temperatures.

New cost-effective method can benefit future 3D printing of metal alloys



An MIT-led team reports a simple, inexpensive way to strengthen a material key to applications in aerospace and nuclear energy generation. The MIT beavers and other shapes in this photo were created using the new technique (Credit: Alexander O'Brien)

Researchers at MIT have discovered a straightforward and cost-effective method to enhance the strength of a crucial material commonly employed in various applications today. Using Inconel 718, a popular 'superalloy,' the team combines it with ceramic nanowires to create a nanoceramic coating. Laser powder bed fusion, a 3D printing method, melts layers

of powder in a specific pattern, creating strong and stretchable parts. The cost-effective process improves thermal resistance and eliminates flaws, benefiting future 3D printing of metal alloys. According to the researchers, the rapid cooling of ultrathin 3D-printed metal alloy layers offers new alloy design possibilities. It surpasses conventional methods, allowing broader compositional exploration. This breakthrough in 3D printing opens up vast possibilities for alloy design and resilient materials in extreme conditions.